Observability and sensor layout for influence-area load monitoring of continuous girder bridges

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Abstract

On a continuous girder the influence line of a channel changes sign over the piers, so the area under a record, which gives the total load in influence-area weighing, can vanish. This paper gives criteria for choosing, before installation, channels that observe the total load. Closed forms give the blind loci of five channel types on continuous and haunched girders; strain gauges are blind at every zero of the uniform-load moment and nearly blind in a band that can contain gauges set back from the piers. Optimal sensors lie at the extremes of the uniform-load response, and the constant follows a linear-fractional (Möbius) law in the scaled pier stiffness. A laboratory test confirms the identity. Public moving-truck records, whose reference weights vary too little to test weight discrimination, are consistent with its assumptions within margins of 10–27 % and show that vehicle-free noise covariance misleads multi-channel weighting. On an in-service box girder, two static readings leave a side-span deflection constant uncertain by a factor of 6.6, whereas the main-span constant is well determined.

Keywords: Structural health monitoring; Sensor placement; Observability; Continuous girder; Bridge weigh-in-motion; Identifiability

1. Introduction

Heavy vehicles above the design envelope shorten the life of highway bridges and, in extreme cases, contribute to collapse (Fan et al., 2015). Knowing the loads that a bridge actually carries is therefore part of risk-informed asset management (Ventura et al., 2024). Pavement

weigh-in-motion (WIM) is mature (Al-Qadi et al., 2016; Jacob et al., 2002), but it weighs the vehicle rather than the load on a particular bridge. Bridge weigh-in-motion (BWIM), introduced by Moses (1979), fits the measured response to axle loads times influence-line ordinates and has been refined with measured influence lines (OBrien et al., 2006), regularisation (OBrien et al., 2009), accuracy improvements (Žnidarič et al., 2018; Yu et al., 2016) and the transverse position of the vehicle (Yu et al., 2018). Moving-force identification recovers axle-load histories (Law et al., 1997; Chan et al., 1999; Law and Zhu, 2000), also on multi-span bridges (Zhu and Law, 2006). These methods aim at axle loads and need several sensors, a structural model or a regularised inversion.

A simpler family estimates only the gross weight from the area under the response. Ojio and Yamada (2002) integrated the strain records of plate-girder stringers and calibrated the result against a truck of known weight; this calibration is called the standard-load method (SLM). It has been used in long-term monitoring (Cardini and DeWolf, 2009) and was accurate to within 5 % on a seven-span slab-on-girder bridge for test trucks with well-estimated speeds (Helmi et al., 2014). Later work used strain areas and integrals (Yang et al., 2021; Zhu et al., 2022; Agüero-Barrantes et al., 2023), shear strain (Lobo Aguilar and Christenson, 2018), sparse sensing that chose gauge number and location on a single-span bridge (Agüero-Barrantes et al., 2021), load-cell reactions (Shi et al., 2025) and an influence-line-free probabilistic index (Yan et al., 2022). Two of the present authors used the integral of mid-span deflection on a simply supported beam (Nguyen et al., 2024).

The identity behind these methods is classical: the area under an influence line equals the response to a uniform unit load, as design codes use for distributed lane loads (Ghali et al., 2017). On continuous girders, however, the influence line changes sign over each interior support, and positive and negative areas partly cancel. Continuous bridges have been weighed from distributed fibre-optic strain (Chaudhary et al., 2025), with lane-specific influence lines (Wu et al., 2020) and from deck strain on long-span box girders (Moghadam et al., 2022, 2023a, 2023b); Deng et al. (2025) replaced the small net area of a multi-span displacement influence line by its absolute area. Sensor placement for BWIM has been posed through an augmented Kalman filter (Zhou et al., 2025), deflection-based load identification as a regularised stochastic problem (Zhang et al., 2026), and deflection influence lines have been inverted for girder stiffness (Zhou et al., 2026). None of these studies says, before installation, which channel at which position can observe the total load at all, and how robustly to uncertain boundary conditions.

This paper supplies such criteria. Neither the influence-area identity nor the generalised least-squares (GLS) estimator is new; the contributions lie in the criteria built on them. (i) We derive a closed-form catalogue of blind loci for five channel types on two- and three-span and haunched girders, with two indicators that need no reference rigidity; at every prismatic location examined, strain influence lines cancel less than deflection influence lines (cancellation ratio $R_{\varepsilon}=0.667$ against $R_{w}=0.571$ at mid-span of two equal spans), yet strain gauges are blind in every continuous girder. (ii) We derive a placement rule, optimal sensors at the extremes of the uniform-load response, and show on one day of public moving-truck records that GLS weights built from vehicle-free noise raise the out-of-sample error above that of the gauge average (20.4 % against 11.9 % of gross weight). (iii) We show that the constant depends on pier stiffness through a linear-fractional (Möbius) law and, on an in-service box girder, that two static readings leave a side-span deflection constant uncertain by a factor of 6.6, while a pier tiltmeter would reduce the Cramér–Rao bound on its logarithm from 0.87 to 0.18.

2. Theory

2.1. Influence-area identity

Consider a linear-elastic structure, unchanged during a crossing, with any distribution $E\,I(x)$ and any linear-elastic supports. A response channel at a fixed point *m* is any fixed linear functional of the displacement field measured from the unloaded state (deflection, rotation, fibre strain $\varepsilon = c\,M/E\,I$, shear force or reaction), with an influence line $\eta_m(x)$ that is zero outside the loaded path $[0, L_p)$. For a convoy of axle loads P_i at fixed offsets d_i ($d_1=0$) crossing quasi-statically, the record is

$$y_m(x)=\sum_{i=1}^{n} P_i\,\eta_m(x-d_i). \tag{1}$$

Its integral over the complete crossing is

$$J_m=\int_{0}^{L_p+d_{max}} y_m\,dx = P_{tot}\,C_m,\quad C_m=\int_{0}^{L_p}\eta_m(u)\,du=\frac{y_m[q_0)}{q_0}, \tag{2}$$

where P_{tot} is the total load and $y_m[q_0)$ the response to a uniform load q_0 over the whole path. This is the classical influence-area identity (Ojio and Yamada, 2002; Ghali et al., 2017); the

second equality needs superposition only. C_m is called the integral constant, or the constant. For a strain gauge at distance c from the neutral axis,

$$C_\varepsilon = \frac{c_m M_q(x_m)}{E I_m q_0}. \tag{3}$$

The section enters only as a gain, so a strain channel is blind exactly at the zeros of the uniform-load moment diagram M_q, for any $EI(x)$ and any elastic supports. The beam-property method (BPM) computes C from a structural model and returns $P = J/C$; SLM takes the constant from a reference crossing of known weight P_0 and returns $P_t = P_0 J_t / J_0$ (Ojio and Yamada, 2002).

Eq. (2) assumes a linear, time-invariant structure, a quasi-static response, axles on the path on which η_m is defined (so each lane has its own constant), a complete crossing and re-zeroing to the unloaded state; Sections 3, 4 and 6 test these assumptions, and truncating the record when the first axle leaves changes the rotation integral by up to 59 % on a two-span example (Supplementary Section S3).

2.2. Closed-form blind loci

For a prismatic span under a uniform load, every channel is fixed by the two end moments, written as $\acute{M} = -8M/(q_0 L^2)$, positive for hogging: $\acute{M}_l$ at the left end and $\acute{M}_r$ at the right end. In the instrumented span of a two-span beam and in the side span of a symmetric three-span beam, $(\acute{M}_l, \acute{M}_r) = (0, \mu)$, with the classical interior-moment parameter (Ghali et al., 2017)

$$\mu_{2span} = \frac{1+\lambda^3/\gamma}{1+\lambda/\gamma}, \mu_{3span} = \frac{2(\gamma+\beta^3)}{2\gamma+3\beta}. \tag{4}$$

Here $\lambda = L_2/L_1$ and $\beta = L_{main}/L_{side}$ are the span ratios and γ is EI_2/EI_1 or EI_{main}/EI_{side} (Fig. 1). The deflection constant at $\xi = x_m/L$ is

$$\frac{192 E I C_m}{L^4} = 8F(\xi,\mu), F(\xi,\mu) = \xi(1-2\xi^2+\xi^3) - \frac{\mu}{2}\xi(1-\xi^2). \tag{5}$$

Its left-hand side is the indicator N_w of Eq. (10). Setting $F = 0$ gives the deflection blind condition, which depends on μ alone:

$$C_m = 0 \Leftrightarrow \mu = \mu_{crit}(\xi) = \frac{2(1+\xi-\xi^2)}{1+\xi}. \tag{6}$$

A deflection blind point therefore exists only for $1<\mu<2$. For a mid-span sensor on two spans of equal rigidity, the blind span ratio and, for equal spans, the optimal sensor position are

$$\lambda_{crit} = \frac{3+\sqrt{33}}{6} = 1.4574\,,\ \xi_{\text{¿}} = \frac{1+\sqrt{33}}{16} = 0.4215\,. \tag{7}$$

For a mid side-span sensor on a symmetric three-span beam of uniform rigidity, the blind ratio is the positive root of $6\beta^3 - 15\beta - 4 = 0$,

$$\beta_{crit} = \sqrt{10/3}\cos\left[1/3\arccos\left(2/5\sqrt{6/5}\right)\right] = 1.7006\,. \tag{8}$$

The locus ends at $\beta = 2\cos(\pi/9) = 1.8794$. A channel is called near-blind when its indicator (Section 2.3) is below 0.2 in magnitude; for this gauge the near-blind band is $1.621<\beta<1.775$ (Fig. 2). Derivations and optimal side-span positions are in Supplementary Section S2 and Table S2, and all closed forms agree with finite element (FE) evaluations of Eq. (2) in 981 checks (Supplementary Section S3).

For strain, with $(\acute{M}_l, \acute{M}_r) = (0, \mu)$, the indicator and the blind point ξ_0 are

$$N_M = \frac{8\,M_q(x_m)}{q_0 L^2} = 4\,\xi(1-\xi) - \acute{M}_l(1-\xi) - \acute{M}_r\xi\,,\ \xi_0 = 1 - \frac{\mu}{4}. \tag{9}$$

Strain blind points exist in every span that has a sagging zone ($\mu<4$), so every continuous girder has them. For two equal spans, $\xi_0 = 0.75$ and the near-blind band $|N_M|<0.2$ lies 0.19–0.32 of the span from the interior support: a gauge at the support section is well conditioned ($|N_M| = \mu$), but one set back from the pier to clear the diaphragm can fall into the band. Table 1 collects the blind loci, and Fig. 3 shows the indicators along two prismatic girders. The end-reaction indicator $N_F = 2F_A/(q_0 L_1)$, with F_A the end reaction, equals ξ_0 for any $EI(x)$ and pier springs, because $M_q = F_A x - q_0 x^2/2$ in the end segment.

Table 1. Blind loci of response channels: positions or span ratios at which the constant C of Eq. (2) vanishes. Prismatic beams with $\gamma = 1$ unless stated; μ from Eq. (4); ζ: position in the main span from the pier, as a fraction of the main span.

Channel	Blind where	Two-span beam (span 1)	Symmetric three-span beam
Strain, moment, curvature	zeros of M_q	$\xi_0 = 1 - \mu/4$ (0.75 at $\lambda = 1$); band $0.676 < \xi < 0.812$ at $\lambda = 1$	side: $\xi_0 = 1 - \mu/4$; main: $\zeta_0 = 1/2\left(1 - \sqrt{1 - \mu/\beta^2}\right)$ from the pier (0.276 at $\beta = 1$)
Deflection	zeros of the uniform-load deflection w_q	$\mu = \mu_{crit}(\xi)$, only for $1 < \mu < 2$; mid-span: $\lambda = 1.4574$, band $1.385 < \lambda < 1.525$	mid side span: $\beta = 1.7006$, band $1.621 < \beta < 1.775$; locus ends at $\beta = 1.8794$
Rotation	extrema of w_q	end: $\mu = 2$ ($\lambda = 1.6180$); pier: $\lambda = 1$ for any γ	end: $\beta = 2\cos(\pi/9)$; pier: $\beta = \sqrt{3/2}$ for any γ; mid main span: always
Shear	maxima of M_q	$\xi = \xi_0/2$	side: $\xi_0/2$; mid main span: always
End reaction	no sagging zone in the end span	$\mu = 4$ ($\lambda = 2.3028$)	$\mu = 4$ ($\beta = 2.6691$)
Interior reaction	never under a full uniform load	–	–

2.3. Observability and conditioning indicators

Every estimator below divides by C, so to first order the sensitivity to measurement error scales as $1/|C|$. For a deflection channel the normalised indicator is

$$N_w = \frac{192\, E\, I_{ref}\, C_m}{L^4}, \tag{10}$$

with L the span containing the sensor and $E\, I_{ref}$ a reference rigidity. N_w equals 2.5 at mid-span of a simply supported span, where N_M and the rotation and shear indicators (Supplementary Section S3) equal 1, so a common threshold of 0.2 is 8 % of the simply supported value for deflection and 20 % for strain; both thresholds are design conventions. On a non-prismatic girder N_w also depends on the choice of $E\, I_{ref}$, which, as Section 5 shows, can dominate the result, whereas N_M, the uniform-load moment relative to $q_0 L^2/8$, needs no reference rigidity. Neither does the cancellation ratio

$$R = \frac{C}{A^{+¿} = 1 - \dfrac{|A^{-})}{A^{+¿}, ¿}} \tag{11}$$

the net influence area divided by its positive part $A^{+¿¿}$. R equals 1 for a simply supported span and 0 at a blind configuration, so a small indicator with R near 1 indicates restraint rather than

cancellation. Strain influence lines cancel less than deflection influence lines at every prismatic location examined: at mid-span of two equal spans, $R_{\varepsilon}=0.667$ against $R_{w}=0.571$.

A residual offset δ_0, in the units of the channel, left after re-zeroing before and after the crossing adds $\delta_0 L_{tot}$ to the integral over a record of length L_{tot}. With $J=C\,P$ the relative bias b of the load is

$$b=\frac{P_{\delta}}{P},\; P_{\delta}=\frac{\delta_0 L_{tot}}{C}. \tag{12}$$

The offset-equivalent load P_{δ}, at which the bias reaches 100 %, combines sensor position, baseline stability and the lightest vehicle to be screened into one number.

2.4. Multi-channel estimation and sensor placement

Let channel j give the integral $J_j=C_j P+\delta_{0,j} L_{tot}+n_j$, with white noise n_j and offsets that may be correlated between channels, and let Σ be the covariance matrix of J. A linear estimator $\hat{P}=a^{\top} J$ is unbiased for every axle configuration if and only if $a^{\top} C=1$, and minimising its variance gives the GLS weights

$$a^{\star}=\frac{\Sigma^{-1} C}{C^{\top}\Sigma^{-1} C},\; Var\,\hat{P}^{\star}=\left(C^{\top}\Sigma^{-1} C\right)^{-1}, \tag{13}$$

the Gauss–Markov best linear unbiased estimator. A non-uniform weighting function along one record cannot do better, because only uniform weighting keeps the estimate independent of the axle configuration for every structure (Supplementary Section S3). Adding the constraint $\sum_j a_j=0$ within each channel type cancels any disturbance shared by the channels, such as an offset, a drift or motion of the reference frame. With equal statistics in all channels, the variance of this zero-sum estimator is

$$Var\,\hat{P}_{zs}=\frac{\sigma^2 \Delta x\, L_{tot}+(1-\rho)\,\delta_0^2 L_{tot}^2}{\sum_j \left(C_j-\acute{C}\right)^2}, \tag{14}$$

with σ the standard deviation of the white noise, Δx the sampling interval and ρ the common-mode fraction of the offset (Supplementary Section S4).

If the noise statistics do not depend on position, $1/Var$ is a convex quadratic in the constants $C(x_j)$ for both Eqs. (13) and (14), and the feasible constants form a box bounded by the smallest and largest $C(x)$ along the girder. A convex function on a box is maximal at a vertex, so optimal sensors lie at the maximum and the minimum of $C(x)$: of $w_q(x)/q_0$ for deflection, with w_q the uniform-load deflection, and of $c\,M_q(x)/(E\,I\,q_0)$ for strain, whose extremes on a haunched girder differ from those of M_q. With independent offsets every sensor should sit near the maximum; with correlated offsets one should sit at the minimum, because a common offset cancels in the difference. A brute-force search found no better design, and a Monte Carlo study confirmed Eqs. (13) and (14) to within 0.6 % (Supplementary Section S4). The zero-sum estimator helps only if the disturbance is common-mode during the crossings themselves, not merely in vehicle-free periods, and Section 4.3 shows that real records can violate this.

2.5. Identifiability of the constant under boundary uncertainty

A girder with monolithic piers has the stiffness matrix $\alpha K_g + k K_p$, where K_g and K_p are its girder and pier parts, α is the ratio of effective to design girder rigidity and k is the rotational stiffness of each pier. Every static response therefore has the form $f(\kappa)/\alpha$, where $\kappa = k/\alpha$ is the scaled pier stiffness. Only the ratio of two static readings, which is independent of α, informs κ. For a mirror-symmetric girder with equal piers, the uniform load excites only the antisymmetric pattern of pier rotations, and the Sherman–Morrison formula gives (Supplementary Section S5)

$$C_m(\alpha,\kappa) = \frac{1}{\alpha}\left[C_\infty + \frac{C_0 - C_\infty}{1+\kappa/k_g}\right]. \tag{15}$$

C_0 and C_∞ are the constants for pinned and for fixed piers at $\alpha = 1$, and k_g is the girder's own stiffness against antisymmetric pier rotation. Eq. (15) is called the Möbius law. The ratio C_∞/C_0 is the full change that pier fixity can produce at a gauge, and the boundary sensitivity $B_m = \left|\partial \ln C_m / \partial \ln \kappa\right)$ is its local slope.

For a symmetric prismatic three-span beam with rotational springs k at the piers, slope-deflection gives, with $\hat{k}$ the pier stiffness relative to the side-span rigidity,

$$\mu(\hat{k}) = 1 + \frac{\beta(2\beta^2 - 3)}{3\beta + 2\gamma + \beta\hat{k}}, \hat{k} = \frac{k\,L_{side}}{E\,I_{side}}. \tag{16}$$

Eq. (16) replaces μ_{3span} in Eq. (5). In the main span the end-moment parameter becomes $\acute{M}_{main} = 2\left(\beta^3 \hat{k} + 3\beta^3 + 3\gamma\right) / \left[3\beta^2\left(\beta \hat{k} + 3\beta + 2\gamma\right)\right]$, which reduces to μ/β^2 for pinned piers. For $\beta = 1.636$, the span ratio of the bridge in Section 5, the side-span indicator at $\xi = 0.4$ rises from 0.287 with pinned piers to 1.037 with fixed piers, while the mid main-span indicator falls from 0.755 to 0.500. Rotational restraint thus moves the side-span gauge away from blindness but makes its constant strongly dependent on k.

3. Laboratory verification

A static test on a two-span beam checks the model and the algorithm, not performance under moving traffic. A steel hollow square tube, 25 mm × 25 mm × 2.03 mm ($I = 16\,530$ mm⁴, $E = 200$ GPa), formed two equal spans of 2500 mm, giving $C_m = l^4/(192\,E\,I) = 61.54$ mm²/N at mid-span of the first span (l the span length). Deflection there was measured with 0.005 mm resolution while calibrated weights of OIML class M1 were suspended at 250 mm steps: a single 40.64 N reference load, two loads 250 mm apart, and three loads. The order of the three loads was not recorded; the order used fits the record best (root-mean-square residual 0.030 mm against 0.55 mm reversed), so the three-load result is not a fully independent test.

The analytical influence line matched the single-load record within 0.0126 mm, 0.44 % of the peak. With the trapezoidal rule, BPM overestimated the weighed totals of 66.99 N and 125.08 N by 1.10 % and 1.19 %, and SLM by 0.61 % and 0.70 % (Supplementary Table S3). The BPM-to-SLM ratio equals 1.0049, the ratio of the measured reference integral to its exact value: SLM absorbs the calibration error of the reference record, whereas BPM does not. The SLM deviations match the growth of the measured-to-analytical integral ratio (1.010, 1.016 and 1.017 for one, two and three loads), which the data cannot attribute to support seating or to the stated masses. The test supports the invariance of Eq. (2) only indirectly, and the predicted blind configuration (spans of 2500 mm and 3644 mm) was not tested.

4. Verification on public moving-truck records

Public records test the assumptions of the identity and the noise model of Section 2.4, with 95 % confidence intervals (CIs); further tables are in Supplementary Section S7.

4.1. Data and processing

The Nothing-on-Road BWIM (NOR-BWIM) dataset (Moghadam and Sarlo, 2024) contains strain records of the Virginia Smart Road Bridge (Moghadam et al., 2022, 2023a, 2023b; Moghadam, 2022), a post-tensioned concrete box girder continuous over five spans of 86.5 + 3 × 144 + 86.5 m = 605 m, from transducers on the top-slab soffit near mid-span of span 2. Day 1 has 12 channels (“ch”), four of them without axle spikes; day 2 has four channels in one line under the wheel path (“st”, for example st36 and st40). On day 1, two 3-axle trucks with driver-reported gross weights ran in the instrumented lane (Moghadam, 2022). On day 2, a 3-axle truck of 26.8 t and a 5-axle truck of 25.9 t, weighed on a static scale (Moghadam, 2022), ran in the instrumented lane (lane A), in the other lane (lane B), and in two runs one behind the other. The weights vary by only a few per cent within each day. Each record was re-zeroed with a linear baseline fitted to vehicle-free segments before and after the crossing, integrated over the whole crossing and converted to $\int \varepsilon \, dx$ with the speed from the axle spikes (Fig. 4); the noise of the integral came from vehicle-free windows of crossing length.

4.2. Invariance, lanes and gauges

Invariance between trucks can be shown only within margins of 10–27 %. On day 1, the χ^2 test does not reject consistency with the measured noise for eight of the twelve gauges, but only ch11 shows equal constants for both trucks within ±10 % (Fig. 5a), about what chance alone would produce with twelve tests at the 5 % level. The four gauges without axle spikes give constants 22–26 % lower for the second truck (CIs exclude 1), but speed and truck are almost perfectly confounded on day 1 (correlation 0.98), so wheel path, local deck action or speed may cause this. On day 2, the ratio of the gauge-average constants of the 5-axle and the 3-axle truck is 0.98 (95 % CI 0.70–1.37; Fig. 5b): consistent with Eq. (2), but with eight runs it cannot exclude differences of up to about 27 %, so it is a coarse check of linearity and processing. A speed effect was neither resolved nor excluded (Supplementary Fig. S11).

Each lane needs its own constant. For the gauge average, return trips in lane B gave constants 0.79 times those of lane A (95 % CI 0.59–1.00), as on another continuous girder (Wu et al., 2020); lane-B speeds came from the lag between gauges, so this ratio also contains any difference between the two speed estimates. The lane effect is clearly resolved for one gauge only (st36: 0.64, 95 % CI 0.47–0.81). For the gauge average, the two runs with the trucks one behind the other gave 0.92 of the summed single-truck predictions (pooled 95 % CI 0.77–1.06), consistent with superposition although both fall below the prediction.

Each deck gauge also needs its own constant. The constant of ch11 is 3.9 times that of ch8, although the gauges are within 6 m of each other and a one-dimensional beam model predicts nearly equal constants, and the st36-to-st40 ratio changes with the lane by about 2.5 standard errors, so gauge-factor spread is an unlikely cause. The deck-soffit gauges also respond to slab bending and shear lag, so the constant must come from a weighed vehicle or a three-dimensional model. A beam model still reproduces the global shape of the strain histories (r^2 = 0.79–0.94) and confirms two points of Section 2: the gauges are not blind, since the zeros of the uniform-load moment diagram in span 2 lie about 35 m away, and the whole 605 m crossing must be integrated, since span 2 contributes about 170 % of the net integral and the adjacent spans −26 % and −66 %.

4.3. Noise model and multi-channel estimators

Single gauges, the gauge average, GLS (Eq. (13)) and the zero-sum estimator (Eq. (14)) were compared by leave-one-run-out cross-validation, with Σ estimated from vehicle-free windows only (Table 2; further estimators and leave-one-truck-out results in Supplementary Table S8 and Fig. S10).

Table 2. Mean absolute error (MAE) of the gross weight (% of gross vehicle weight), leave-one-run-out, with 95 % bootstrap CIs over the test runs (n = 8 on day 2, n = 7 on day 1). GLS uses Σ measured in vehicle-free windows.

Estimator	Day 2, 4 gauges	Day 1, 12 gauges
Gauge average	11.9 (6.0–18.6)	5.3 (2.5–7.8)
GLS, measured Σ	20.4 (12.3–28.8)	5.8 (3.5–8.1)
Best single gauge (chosen after the fact)	8.4 (2.9–15.3)	4.8 (1.8–7.9)
Mean weight of the calibration runs	2.0 (1.9–2.0)	2.2 (1.1–3.5)

On day 2, GLS was worse than the gauge average, 20.4 % against 11.9 % MAE, with a paired difference of +8.5 percentage points (95 % CI +2.3 to +15.1); the zero-sum estimator (20.3 %) was worse as well. On day 1 the estimators cannot be told apart. Because the reference weights hardly vary, a predictor that always returns the mean calibration weight is better than every estimator on both days, and the records cannot show weight discrimination.

The failure of GLS is itself a finding. The vehicle-free noise of the four day-2 channels is strongly common-mode, but the crossing residuals (deviations of each crossing's integral from its calibrated value) are much less correlated between gauges: 0.63 against 0.90 for vehicle-free noise at the same window length. The vehicle-free covariance therefore predicted

GLS to beat the average, whereas the residual covariance predicts the observed reverse order (Supplementary Section S7). The gauge-specific component of each crossing, plausibly from the wheel path and local slab action, is amplified by the large GLS weights of opposite sign on st40 and st36, and estimating Σ from the residuals over-fits with only eight runs. Σ, and hence ρ, should be estimated from the residuals of enough calibration crossings, not from vehicle-free windows alone.

5. Design case: an in-service three-span box girder

5.1. Bridge, data and model

The Cầu Mây Bridge crosses the Kinh Môn River in Hải Dương, Viet Nam. Its main bridge is a 55 + 90 + 55 m continuous prestressed concrete box girder built by balanced cantilever ($\beta = 1.636$), 2.40 m deep at mid-span and 5.40 m at the piers. The inner piers, including T7, are monolithic with the girder; the end supports T5 and T8 carry pot bearings. In the June 2020 acceptance test, 3-axle trucks with a weighed mean of 25.06 t stood in the side and main spans, and dial gauges read the peak deflection at N8, in the side span at 0.4L (22 m) from T8, and at N7, mid main span. Each section has six readings (three points across, central and eccentric arrangements), with means of 5.55 mm at N8 and 15.42 mm at N7 (Supplementary Tables S11–S13).

The Euler–Bernoulli model has a parabolic depth variation over $L_v = 44$ m from each inner pier, a second moment proportional to $h^{2.943}$, where h is the girder depth, and a rotational spring k at each inner pier; other haunch lengths and exponents and a rigid pier zone were tested as model-form variants (Supplementary Section S8). Calibration A, the design model, has $\alpha = 1$ and $k = 2.75 \times 10^7$ kN m/rad fitted to the designer's FE output and deviates from the measurements by +2.4 % at N8 and +12.3 % at N7. Calibration B, the joint least-squares fit, has $\alpha = 0.716$ and $k = 1.2 \times 10^8$ kN m/rad, with misfits of −2.8 % and +2.9 %. A lies at $\kappa = 2.75 \times 10^7$ kN m/rad and B at $\kappa = 1.68 \times 10^8$ kN m/rad. Fig. 6 shows the indicators along the girder.

5.2. What the static data identify

The static data leave the constant of the side-span gauge N8 uncertain by a factor of 6.6, whereas that of the main-span gauge N7 is well determined, because of a model-form misfit and the Möbius law of Eq. (15).

The test quantity r_{87}, the measured-to-model ratio at N8 divided by that at N7, would equal 1 if one model fitted both gauges. Its minimum, 1.058, lies at calibration B (Fig. 7a), and the model-form variants move the minimum between 1.039 and 1.092 but never to 1. The 4–9 % residual is therefore model-form error, calibration B is the nearest fit rather than an exact one, and its $\alpha = 0.716$ would mean a 28 % loss of rigidity in a recently built prestressed girder.

The admissible set contains every model whose misfit at each gauge stays within a tolerance τ. The six readings of each section have a standard deviation of 4.7–5.0 % about their mean, so the standard error of the mean is 1.9–2.0 %; this scatter comes from the transverse positions, not from repeatability, because the readings are not replicates. Combined in quadrature with a model-form standard uncertainty of 1.75 %, it gives 2.64 % for each section mean and $\tau = 5.3$ % at two standard deviations. At this tolerance the data bound κ only from below: every $\kappa \geq 1.8 \times 10^7$ kN m/rad is admissible, with α between 0.51 and 1.15. At $\tau = 3$ % the admissible set shrinks and excludes the design rigidity, $\alpha = 1$ (Supplementary Table S10).

Over the admissible set, C_8 ranges from 3.0×10^{-5} to 2.0×10^{-4} m²/kN, whereas C_7 stays within 3.3–5.0 × 10^{-4} m²/kN; the N8 indicator spans 0.124–0.822 against 0.190–0.286 for N7 (Table 3). Going from pinned to fixed piers multiplies the N8 constant by 14.0 but reduces the N7 constant to 0.59 of its pinned value, and the design stiffness sits at $\kappa / k_g = 0.63$, near the steepest part of the Möbius curve, where B_m is 0.51 at N8 and 0.12 at N7. Calibrating to the N8 reading does not fix the N8 constant either: the ratio of C_m to the modelled N8 deflection varies by a factor of 6.5 over the admissible set, against 1.5 for N7.

Table 3. Deflection indicator N_w, Eq. (10), and cancellation ratio R, Eq. (11), of the installed Cầu Mây gauges; EI_{ref} = design EI_{mid} (mid main span), L = span containing the gauge. $\alpha = 1$ except for calibration B ($\alpha = 0.716$) and the admissible set (τ = 5.3 %, α = 0.51–1.15); haunched pinned-pier and fixed-pier rows give the extremes over all model-form variants. Values at mid side span: Supplementary Table S14. –: not computed.

Model	N_w, N8 at 0.4L	N_w, N7	R, N8 at 0.4L	R, N7
Prismatic beam, pinned piers	0.287	0.755	0.14	0.70
Haunched, calibration A	0.181	0.209	0.18	0.59
Haunched, calibration B	0.476[a]	0.234	0.56	0.77
Haunched, admissible set	0.124–0.822[b]	0.190–0.286	0.13–1.00	0.56–1.00
Haunched, pinned piers	0.023	0.263	–	–
Haunched, fixed piers	0.446	0.145	–	–

[a] Normalised by the effective rigidity $\alpha E I_{mid}$: 0.341. [b] The upper part requires $\alpha \approx 0.5$; normalised by the effective rigidity: 0.144–0.422.

The main-span indicator is low (0.19–0.29) although R is 0.56–1.00, so its smallness reflects restraint by the haunches, not cancellation.

5.3. Which channel, where

For this girder the deflection channel at mid main span is the robust choice. A side-span deflection gauge is ill-conditioned with respect to pier stiffness and should not take its constant from a design model unless that stiffness is identified.

Strain channels behave differently (Fig. 6). The haunches move every zero-moment point towards mid-span: the side-span zero lies 26.0–33.3 m from the end support, so a strain gauge at mid side span would be practically blind under calibration A ($N_M = 0.048$), and mid main span lies in or at the edge of the near-blind band in every haunched model (N_M = 0.163–0.212 at N7), again because of restraint (R_ε = 0.65–1). By N_M alone the best strain locations in all haunched models are 5–10 m from the piers ($|N_M| \geq 0.39$), but with $c \propto h$ their gain c/EI is about one fifth of that at mid-span (Fig. 6c), so the placement rule, which ranks by $C_\varepsilon = c\, M_q / (E I\, q_0)$, disagrees; which ranking governs depends on whether the strain resolution or the offset sets the noise. Load cells under the end bearings ($N_F = \xi_0$ = 0.47–0.61) cannot be blind unless the side span loses its sagging zone.

The placement rule puts a second deflection sensor at the most negative constant, in the side span 4–10 m from the inner pier; the pair at 100 m and 152 m from T5 is within 0.9 % of the calibration-specific optimum under both A and B, so placement is insensitive to the unresolved pier stiffness. Under the assumed noise model ($\sigma = 0.01$ mm, $\delta_0 = 0.1$ mm, a 220 m record, $\rho = 0.9$, calibration A), the standard deviation of the load falls from 24.8 % with N7 alone to 10.6 % with the pair; this gain is an upper bound, because such common-mode behaviour need not hold during crossings (Section 4.3).

If a model-based constant is wanted, a tiltmeter on the monolithic pier T7 is the most informative additional measurement; under the side-span convoy it would read about 167 µrad. At calibration A ($\kappa = 2.75 \times 10^7$ kN m/rad, α refitted to $\alpha' = 1.07$; Fig. 7b), with 2.64 % standard uncertainty per reading and an assumed 3 % for the rotation, the Cramér–Rao bound on the standard deviation of $\ln C_8$ is 0.87 from the two deflections and 0.18 with the tiltmeter.

The first and third vertical frequencies would give 0.24 instead, but only if the first frequency were identified, and the test reports one frequency per span (Supplementary Table S13). At calibration B the two-reading Fisher matrix is singular; with the tiltmeter the bound is 0.04.

A moving-load test with weighed trucks has been designed for this bridge but not performed. Table 4 gives the predicted values that would decide between the calibrations; the ratio of integrals at the side-span and main-span gauges is independent of truck weight and absolute rigidity, and it differs by a factor of 2.3 between A and B.

Table 4. Predicted discriminating measurements at Cầu Mây for a 25 t 3-axle truck (predictions, not measured).

Measured quantity	Calibration A	Calibration B
Integral ratio, N8 at 0.4L / N7 at mid-span	0.121	0.284
Peak girder rotation over pier T7 (µrad)	−58	−29
Strain integral at the zero-moment point of A (171.3 m) / at N7	0.006	0.17

6. Design implications and limitations

For design before installation (checklist in Supplementary Section S10), one static analysis under a uniform unit load on each lane gives the constant of every candidate channel. Strain gauges should stay out of the near-blind band $|N_M| < 0.2$ around every zero-moment point, including gauges set back from a pier, and deflection gauges out of the bands of Table 1. Eq. (15) gives the constant over the plausible range of pier stiffness; locations with large C_∞ / C_0 or B_m should be avoided, or a measurement that identifies κ planned (a pier tiltmeter, an identified first frequency or an integral ratio between two gauges). Sensors belong at the extremes of $C(x)$. SLM with weighed vehicles of clearly different weights and lane-specific constants is preferred; GLS or zero-sum weights need Σ from calibration-crossing residuals, checked out of sample.

The residual offset limits single-channel accuracy on long spans. Over the Cầu Mây admissible set, with $L_{tot} = 200$ m, $\delta_0 = 0.1$ mm biases a 25 t truck by 16–25 % at a single main-span gauge and by 41–271 % at a single side-span gauge (Supplementary Table S15). With present absolute-deflection technology, single-point integral weighing on a long-span girder is therefore marginal even at the best location, and non-contact techniques must establish their residual offset on site, because environmental effects act over the same window (Sohn, 2007; Cross et al., 2013). Speed enters through the response and its measurement: on

lane A of the field records, speeds from a low-pass fit were 1.139 times those from the axle spikes (standard deviation 0.124, 22 runs), so the speed estimator alone can shift the weight by more than 10 %. The quasi-static error for a constant force crossing the laboratory beam is at most about 1 % when $t_{cr}/T_1 \geq 8$ (crossing time over fundamental period; Frýba, 1999); the bound omits vehicle–bridge interaction and road roughness (Supplementary Sections S9 and S10). The method is therefore proposed for preliminary overload screening, with the threshold set with a margin below the legal limit and the design live loads (CEN, 2003; AASHTO, 2024) and instrumented BWIM for flagged vehicles; influence-line BWIM does not suffer the blindness of Table 1, because it fits the whole record shape.

Several limitations remain. The laboratory beam had equal spans and statically positioned loads, and the blind configuration was not tested. The field data come from one public dataset with four trucks, nearly constant reference weights and no gauge at a zero-moment point, so weight discrimination, the deflection blind loci and a deflection pair could not be tested, and a screening threshold still needs field calibration with trucks of clearly different weights, which this study did not have. The in-service analysis uses static data and a beam model with an assumed depth law and equal pier springs, and its moving-load test is designed but not performed. The treatment is one-dimensional along each lane.

7. Conclusions

1. Strain channels are blind at every zero of the uniform-load moment diagram, which every continuous girder has, and nearly blind 0.19–0.32 of the span from the interior support on two equal spans. Deflection channels are blind only near particular span ratios, such as $\lambda = (3+\sqrt{33})/6$ for two spans and $\beta = 1.7006$ for a mid side-span gauge on three symmetric spans.

2. The indicator N_M and the cancellation ratio R need no reference rigidity, and R separates cancellation from restraint. Optimal sensors lie at the extremes of the uniform-load response, for strain at the extremes of $c\,M_q/EI$.

3. The constant is a linear-fractional (Möbius) function of the scaled pier stiffness. On a 55 + 90 + 55 m box girder, two static readings leave the side-span deflection constant uncertain by a factor of 6.6, whereas the main-span gauge is well determined; a pier tiltmeter would reduce the Cramér–Rao bound on $\ln C_8$ from 0.87 to 0.18.

4. On public moving-truck records, the constants were consistent with invariance and superposition within the 10–27 % that the noise resolves, and each lane and deck gauge needed its own constant. GLS weights from vehicle-free noise were worse out of sample than the gauge average (20.4 % against 11.9 %), because crossing residuals were much less correlated between gauges (0.63 against 0.90).

5. On that girder, with a 200 m record, a residual offset of 0.1 mm would bias a 25 t truck by 16–25 % even at mid main span.

The next steps are the moving-load test designed for the in-service girder, laboratory tests of the blind configuration, and field calibration with trucks of clearly different weights.

Disclosure of Generative AI Use

During the preparation of this work the authors used Claude (Anthropic) to assist with code development and debugging and with language editing. All scientific formulations, procedures, numerical results and interpretations were independently reviewed and verified by the authors, who take full responsibility for the content of the manuscript.

Acknowledgements

The authors gratefully acknowledge the Laboratory of Transportation Technology and Systems, Faculty of Bridge and Road Engineering, Hanoi University of Civil Engineering, for providing the test facilities used in this study, and A. Moghadam and R. Sarlo for making the NOR-BWIM dataset publicly available. They also thank the Hải Dương provincial project management unit and Công ty TNHH Tư vấn Đại học Xây dựng for permission to use the acceptance-test data of the Cầu Mây Bridge.

Author contributions

Quoc-Bao Nguyen: Conceptualization, Methodology, Formal analysis, Software, Writing – original draft. Thai-Son Vu: Conceptualization, Investigation, Validation, Supervision, Project administration, Writing – review & editing. Hung-Son Nguyen: Investigation, Data curation, Visualization, Writing – review & editing.

Statements and Declarations

Ethical considerations

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interest

The Cầu Mây load test analysed in Section 5 was performed by an author of this paper (Q.-B. Nguyen) as part of inspection work carried out by Công ty TNHH Tư vấn Đại học Xây dựng (HUCE consultancy). The authors declare that there is no other conflict of interest regarding the publication of this article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors. The work was carried out as part of the authors' employment at Hanoi University of Civil Engineering.

Data availability

The field strain records analysed in Section 4 are publicly available in the NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0; Moghadam and Sarlo, 2024). The Cầu Mây load test was performed by an author of this paper, and the authors are permitted to publish the processed data. The laboratory and Cầu Mây data, the analysis scripts with the console output of each run, and the derived data behind every figure and table are provided as Supplementary Material. Every equation and numerical output was checked independently against an analytical, finite element or convergence-based route.

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Figures

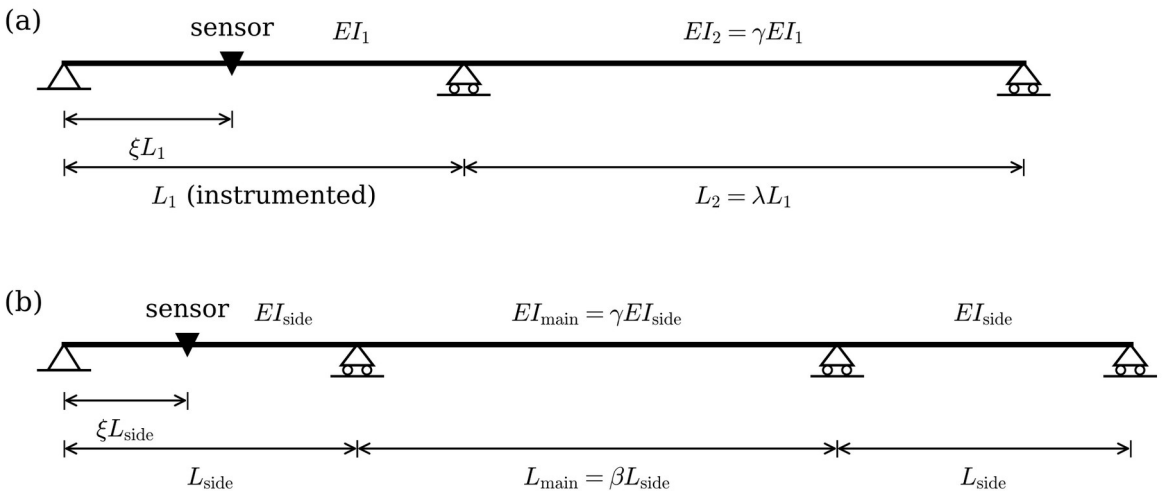


Figure 1. Continuous beams considered: (a) two-span beam with instrumented span L_1 of rigidity EI_1, adjacent span $L_2 = \lambda L_1$ of rigidity $EI_2 = \gamma EI_1$, and sensor at ξL_1; (b) symmetric three-span beam with side spans L_{side} of rigidity EI_{side}, main span $L_{main} = \beta L_{side}$ of rigidity $EI_{main} = \gamma EI_{side}$, and side-span sensor at ξL_{side}.

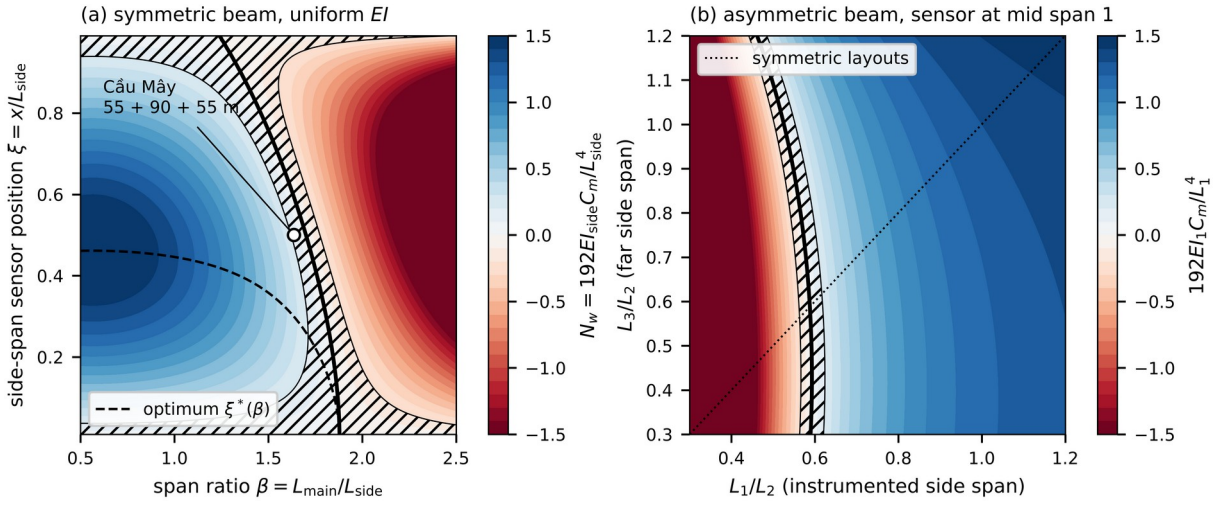


Figure 2. Deflection observability of three-span beams of uniform rigidity: (a) symmetric beam, N_w over β and side-span sensor position ξ, with blind locus (thick line), near-blind band $|N_w| < 0.2$ (hatched), optimum $\xi^*(\beta)$ (dashed) and Cầu Mây proportions with a mid side-span sensor (circle); (b) asymmetric beam, sensor at mid-span of span 1, over L_1/L_2 and L_3/L_2.

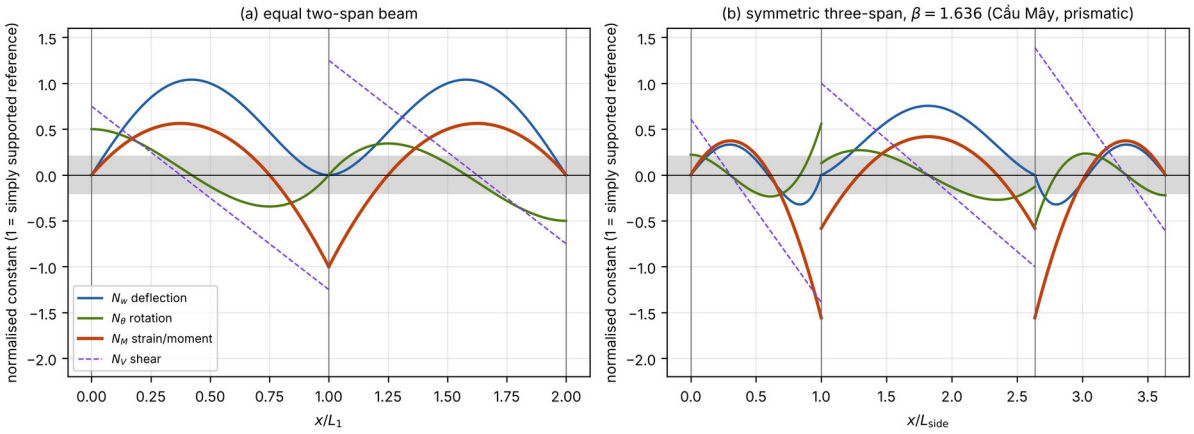
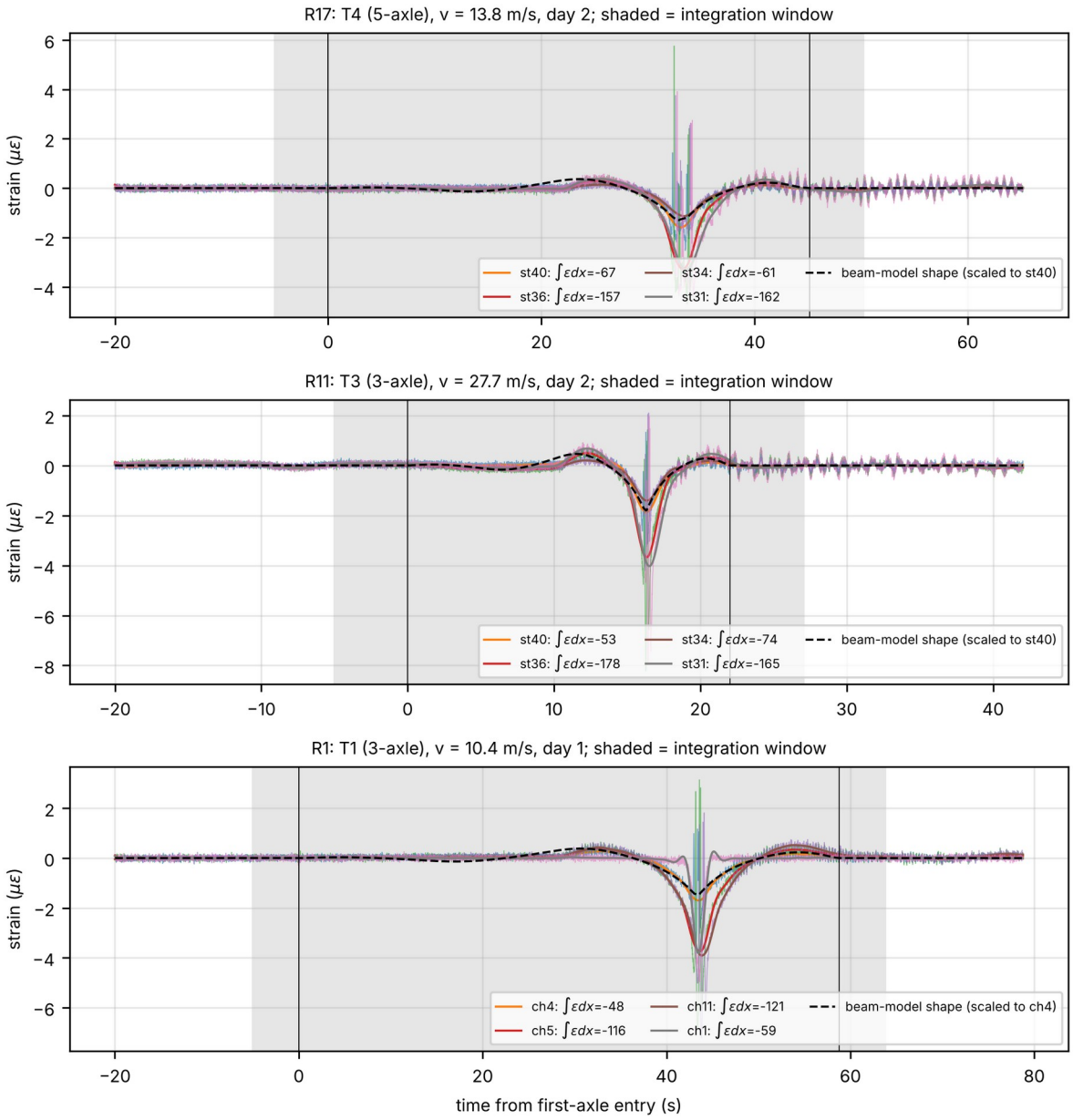


Figure 3. Indicators along prismatic girders: deflection N_w (2.5 at a simply supported mid-span), rotation N_θ, strain N_M and shear N_V (1 for the simply supported reference). (a) Equal two-span beam; (b) symmetric three-span beam, $\beta = 1.636$. Grey band: $|N| < 0.2$.



595

596 **Figure 4.** NOR-BWIM strain records of three single-truck crossings of the five-span girder.
597 Shaded: integration window; dashed: beam-model shape; legend: integral $\int \varepsilon \, dx$ (με m) per
598 gauge.

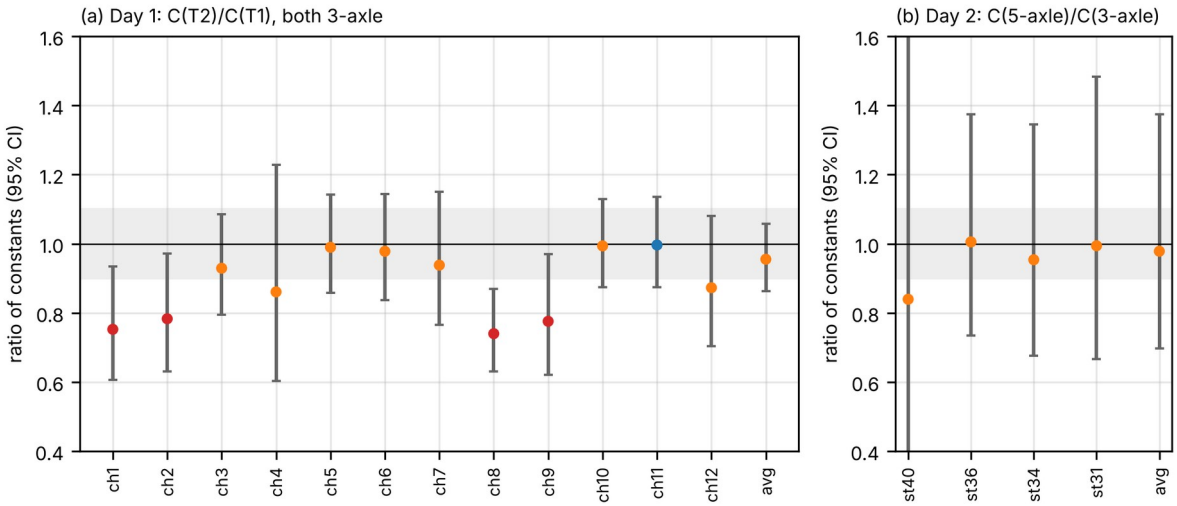


Figure 5. Ratio of the integral constants of two trucks, per gauge and gauge average, with 95 % confidence intervals: (a) day 1, second to first 3-axle truck, gauge average over the four wheel-path gauges; (b) day 2, 5-axle to 3-axle truck. Grey bands: ±5 % and ±10 %. Red: invariance rejected; blue: equivalent within ±10 %; orange: inconclusive.

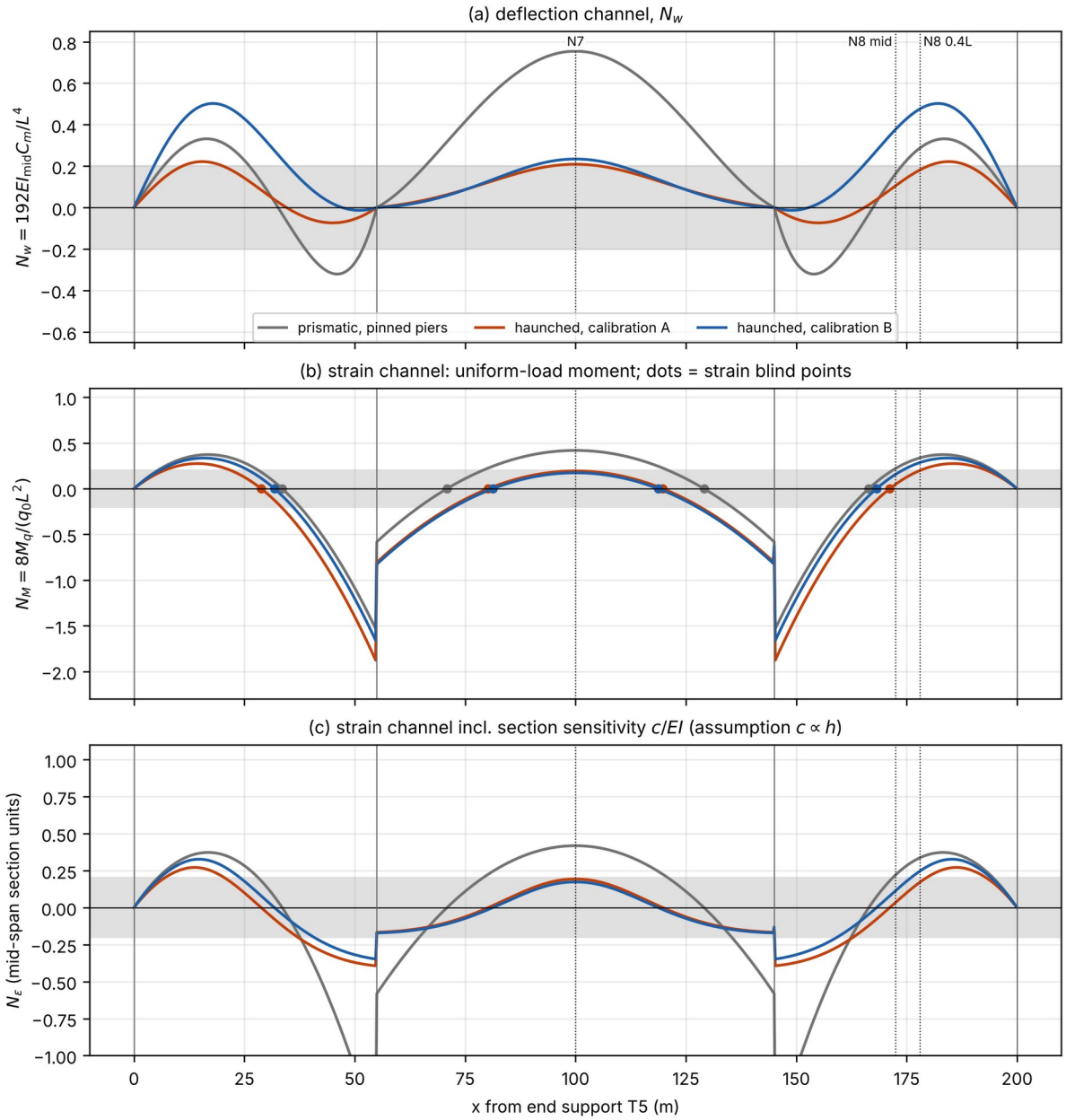
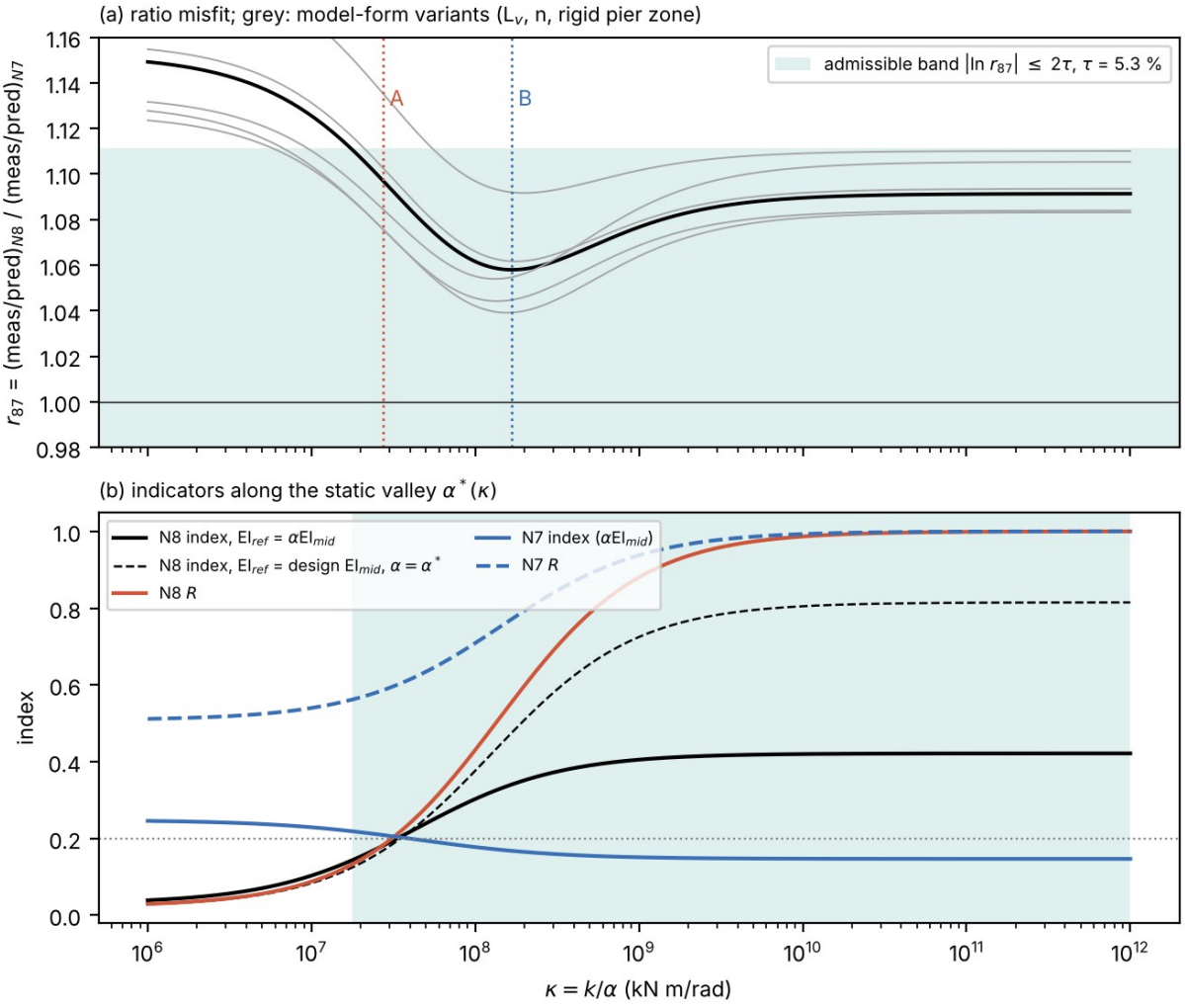


Figure 6. Cầu Mây girder: indicators for a prismatic model with pinned piers and for the haunched model under calibrations A and B: (a) deflection N_w; (b) strain N_M; (c) strain indicator including the gain c/EI, with $c \propto h$. Dotted lines: gauge positions.



608

609 **Figure 7.** Identifiability of the Cầu Mây pier stiffness: (a) misfit ratio r_{87} against $\kappa = k/\alpha$, with
610 model-form variants (grey) and admissible band $|\ln r_{87}| \le 2\tau$, τ = 5.3 %; (b) N8 and N7
611 indicators and cancellation ratios along the best static fit $\alpha^*(\kappa)$; shaded: admissible range of κ.